

Mathematical Properties of the Affine-DDPNM

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1 Preliminaries and notation

1.1 Domain, interfaces, and subdomains

The pore domain $\Omega \subset \mathbb{R}^3$ is partitioned into N non-overlapping Lipschitz subdomains $\{\Omega_i\}_{i=1}^N$ with m **planar** internal interfaces $\{\gamma_k\}_{k=1}^m$, $\gamma_k = \partial\Omega_i \cap \partial\Omega_j$. Each Ω_i has m_i ports, m_i^u internal and m_i^k boundary. We assume every Ω_i possesses a positive-measure no-slip wall segment; this, together with the discrete inf-sup condition of the Taylor–Hood $[P_2]^3$ – P_1 pair on a shape-regular tetrahedral mesh, guarantees that the local saddle-point matrix K_i is invertible.

1.2 Sign convention and interface traction

On each interface γ_k , fix an orthonormal frame $\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$ with $\mathbf{n}_k$ pointing from pore i to pore j (where $i < j$ in the interface pair). For pore i , the outward normal is $\mathbf{n}_{i,k} = \sigma_{i,k} \mathbf{n}_k$ with $\sigma_{i,k} = +1$ if i is the first member and -1 otherwise. The **signed direction vectors** are

$$\mathbf{d}_{i,k,\mathbf{n}} = \mathbf{n}_{i,k}, \quad \mathbf{d}_{i,k,\mathbf{t}_\nu} = \sigma_{i,k} \mathbf{t}_{k,\nu} \quad (\nu = 1, 2), \quad (1)$$

satisfying $\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta}$. We abbreviate $\mathbf{d}_{i,\gamma,\beta}$ when the context is clear.

Traction Ansatz. Using the pseudostress $\boldsymbol{\sigma}(\mathbf{u}, p) := -p\mathbf{I} + \nu\nabla\mathbf{u}$, the applied traction on an internal interface γ as seen from Ω_i is

$$-\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n}_{i,\gamma} = \sum_{(\alpha,\beta) \in \mathcal{M}} \lambda_{\gamma,\alpha,\beta} \varphi_\alpha(s, t) \mathbf{d}_{i,\gamma,\beta}, \quad (2)$$

where $\mathcal{M} = \{0, s, t\} \times \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$, $\varphi_0 = 1$, $\varphi_s = s$, $\varphi_t = t$, and s, t are dimensionless half-span-normalised in-plane coordinates. The minus sign on the left conforms to the classic notation $-\sigma n = p_\gamma n$, so that a positive $\lambda_{\gamma,0,\mathbf{n}}$ corresponds to an isotropic pressure acting into the pore. Equation (2) implies $\boldsymbol{\sigma} \mathbf{n}_{i,\gamma} = -\sum \lambda \varphi \mathbf{d}_{i,\gamma,\beta}$; this is the sign that enters the boundary term of the weak formulation.

For inlet/outlet ports the same convention (2) is used with the single mode $(\alpha, \beta) = (0, \mathbf{n})$ and a prescribed coefficient equal to the boundary pressure.

1.3 Unit loads and local compliance map

For each mode $(\alpha, \beta) \in \mathcal{M}$ on an internal interface γ of Ω_i , the Neumann load vector is

$$[\mathbf{f}_{i,\gamma}^{(\alpha,\beta)}]_j = \int_\gamma \varphi_\alpha(s, t) \mathbf{d}_{i,\gamma,\beta} \cdot \boldsymbol{\phi}_j^{(i)} \, dS, \quad (3)$$

where $\{\boldsymbol{\phi}_j^{(i)}\}$ is the local $[P_2]^3$ velocity basis. There is no leading minus sign: from (2) with $\lambda_{\gamma,\alpha,\beta} = 1$ (all others zero), the applied boundary force is $+\mathbf{d}_{i,\gamma,\beta} \varphi_\alpha$, which is exactly the integrand of (3). For boundary ports only $(\alpha, \beta) = (0, \mathbf{n})$ is used with a prescribed coefficient.

The discrete local Stokes system (no pressure stabilisation) reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{i,\gamma}^{(\alpha,\beta)} \\ \mathbf{X}_{i,\gamma}^{(\alpha,\beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,\gamma}^{(\alpha,\beta)} \\ \mathbf{0} \end{bmatrix}. \quad (4)$$

Let $n_i = 9m_i^u + m_i^k$ be the number of local traction coefficients (the m_i^k boundary entries are prescribed, not unknown). Define $L_i \in \mathbb{R}^{n_i, u \times n_i}$ as the matrix whose columns are all $\mathbf{f}_i^{(\mu)}$, μ ranging over the n_i modes. The local compliance (Neumann-to-Dirichlet) map is

$$\mathbf{G}_i = L_i^\top H_i L_i \in \mathbb{R}^{n_i \times n_i}, \quad H_i = A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}, \quad M_i = B_i A_i^{-1} B_i^\top. \quad (5)$$

$\mathbf{G}_i$ is symmetric positive semi-definite (Lemma 2.2). It is partitioned as

$$\mathbf{G}_i = \begin{bmatrix} \mathbf{G}_i^{uu} & \mathbf{G}_i^{uk} \\ \mathbf{G}_i^{ku} & \mathbf{G}_i^{kk} \end{bmatrix},$$

where $\mathbf{G}_i^{uu} \in \mathbb{R}^{9m_i^u \times 9m_i^u}$ couples internal modes and $\mathbf{G}_i^{uk} \in \mathbb{R}^{9m_i^u \times m_i^k}$ couples internal to boundary.

1.4 Weighted velocity moments and global Schur complement

For mode (α, β) on interface γ of Ω_i ,

$$M_{i,\gamma}^{(\alpha,\beta)} = \int_\gamma \varphi_\alpha(s, t) \mathbf{u}_{i,h} \cdot \mathbf{d}_{i,\gamma,\beta} \, dS. \quad (6)$$

Only $M_{i,\gamma}^{(0,\mathbf{n})} = \int_\gamma \mathbf{u} \cdot \mathbf{n} \, dS$ is the volumetric flow rate; the other eight are weighted velocity moments. From (1), $M_{i,\gamma}^{(\alpha,\beta)} + M_{j,\gamma}^{(\alpha,\beta)} = 0$ for any continuous velocity field.

Enforcing continuity of all nine weighted velocity moments on every internal interface yields the global Schur system

$$\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}, \quad \mathbf{S} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k, \quad (7)$$

where $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$, $\boldsymbol{\lambda}^k \in \mathbb{R}^{m^k}$, and R_i^u, R_i^k are Boolean restriction maps.

2 Mathematical properties

Lemma 2.1 (Discrete incompressibility). $B_i \mathbf{U}_{i,\gamma}^{(\alpha,\beta)} = \mathbf{0}$ for every unit response.

Proof. From the second block row of (4). The explicit expression $\mathbf{U}_{i,\gamma}^{(\alpha,\beta)} = H_i \mathbf{f}_{i,\gamma}^{(\alpha,\beta)}$ follows from standard Schur-complement algebra; the verification $H_i^\top A_i H_i = H_i$ and the consequence $B_i H_i = 0$ are obtained by direct expansion (see the companion methodology document). $\square$

Lemma 2.2 (Symmetry and energy identity). $\mathbf{G}_i$ is symmetric positive semi-definite and $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = \mathbf{U}_i^\top A_i \mathbf{U}_i \geq 0$.

Proof. $\mathbf{G}_i(\mu, \mu') = [\mathbf{f}_i^{(\mu)}]^\top H_i \mathbf{f}_i^{(\mu')} = [\mathbf{U}_i^{(\mu)}]^\top A_i \mathbf{U}_i^{(\mu')}$, symmetric by symmetry of A_i . Summing with coefficients $\lambda_{i,\mu}$ gives the energy identity; $A_i \succ 0$ implies $\mathbf{G}_i \succeq 0$. $\square$

Lemma 2.3 (Kernel of the local compliance map). Let $\mathbf{1}_{(0,\mathbf{n})} \in \mathbb{R}^{n_i}$ have entry 1 at every mode $(\gamma, 0, \mathbf{n})$ (internal and boundary) and 0 elsewhere. Assume

$$\text{rank}(H_i^{1/2} L_i) = n_i - 1. \quad (8)$$

Then $\ker(\mathbf{G}_i) = \text{span}\{\mathbf{1}_{(0,\mathbf{n})}\}$ is exactly one-dimensional.

Proof. (i) $\mathbf{1}_{(0,\mathbf{n})} \in \ker(\mathbf{G}_i)$. Summing the constant-normal loads and using the divergence theorem, $\sum_{\gamma} \mathbf{f}_i^{(\gamma,0,\mathbf{n})} = B_i^{\top} \mathbf{1}$ (coefficient vector of $p_h \equiv 1$). Hence $(\mathbf{U}_i, \mathbf{X}_i) = (\mathbf{0}, p_{\text{const}} \mathbf{1})$ solves (4) for uniform $\lambda_{\gamma,0,\mathbf{n}} \equiv p_{\text{const}}$. By uniqueness, $\mathbf{U}_i = \mathbf{0}$, so $\mathbf{G}_i \mathbf{1}_{(0,\mathbf{n})} = L_i^{\top} H_i L_i \mathbf{1}_{(0,\mathbf{n})} = L_i^{\top} H_i^{1/2} (H_i^{1/2} L_i \mathbf{1}_{(0,\mathbf{n})}) = \mathbf{0}$.

(ii) **Dimension of the kernel.** $\mathbf{G}_i = (H_i^{1/2} L_i)^{\top} (H_i^{1/2} L_i)$, hence $\ker(\mathbf{G}_i) = \ker(H_i^{1/2} L_i)$. **Note:** H_i is not SPD on $\text{Range}(L_i)$; indeed $\text{Range}(L_i) \ni B_i^{\top} \mathbf{1} \in \ker(H_i)$. The equality $\ker(L_i^{\top} H_i L_i) = \ker(H_i^{1/2} L_i)$ holds because $H_i \succeq 0$, so $x^{\top} L_i^{\top} H_i L_i x = \|H_i^{1/2} L_i x\|^2$, which vanishes iff $H_i^{1/2} L_i x = 0$. By (8), $\dim \ker(H_i^{1/2} L_i) = n_i - (n_i - 1) = 1$. Part (i) provides one non-zero kernel vector, hence $\ker(\mathbf{G}_i) = \text{span}\{\mathbf{1}_{(0,\mathbf{n})}\}$. $\square$

Remark 2.4. Assumption (8) states that the n_i projected responses $H_i^{1/2} \mathbf{f}_i^{(\mu)}$ satisfy exactly one linear relation—the constant-pressure identity $\sum_{\gamma} H_i^{1/2} \mathbf{f}_i^{(\gamma,0,\mathbf{n})} = \mathbf{0}$. This is verified numerically for all tested geometries (each G_i has exactly one zero eigenvalue to machine precision). The rank condition concerns only the n_i projected responses, not the much larger space $\text{Range}(H_i)$ of all divergence-free velocities; the codimension of $\text{Range}(H_i^{1/2} L_i)$ in $\text{Range}(H_i^{1/2})$ is > 1 in general and irrelevant to the argument. A rigorous proof of (8) for general Lipschitz pores is open; the numerical evidence is uniform across all tested configurations.

Theorem 2.5 (Solvability and mass conservation). Assume Lemma 2.3 holds for every pore. Further assume:

- (A1) Every connected component of the pore adjacency graph contains at least one pore with a prescribed-traction (inlet/outlet) port (global anchoring);
- (A2) Each pore has a positive-measure no-slip wall segment (local coercivity).

Then $\mathbf{S}$ is SPD, the system $\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}$ has a unique solution, and the reconstructed velocity satisfies local and global discrete mass conservation ($\sum_r M_{i,r}^{(0,\mathbf{n})} = 0$ for each pore, $\int_{\partial\Omega} \mathbf{u}_h^{DD} \cdot \mathbf{n} \, dS = 0$).

Proof. **SPD.** For $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$, let $\boldsymbol{\lambda}_i^u = R_i^u \boldsymbol{\lambda}$. Then $\boldsymbol{\lambda}^{\top} \mathbf{S} \boldsymbol{\lambda} = \sum_i (\boldsymbol{\lambda}_i^u)^{\top} \mathbf{G}_i^{uu} \boldsymbol{\lambda}_i^u = \sum_i \mathbf{U}_i^{\top} A_i \mathbf{U}_i \geq 0$.

If $\boldsymbol{\lambda}^{\top} \mathbf{S} \boldsymbol{\lambda} = 0$, then $\mathbf{U}_i = \mathbf{0}$ for all i . **Zero-extension argument.** Define $\bar{\boldsymbol{\lambda}}_i = (\boldsymbol{\lambda}_i^u; \mathbf{0}) \in \mathbb{R}^{n_i}$, i.e. extend $\boldsymbol{\lambda}_i^u$ by zeros on the boundary ports. Since $(\boldsymbol{\lambda}_i^u)^{\top} \mathbf{G}_i^{uu} \boldsymbol{\lambda}_i^u = \bar{\boldsymbol{\lambda}}_i^{\top} \mathbf{G}_i \bar{\boldsymbol{\lambda}}_i = 0$, Lemma 2.3 gives $\bar{\boldsymbol{\lambda}}_i = c_i \mathbf{1}_{(0,\mathbf{n})}$ for some $c_i \in \mathbb{R}$.

If Ω_i has a boundary port, the corresponding entry of $\bar{\boldsymbol{\lambda}}_i$ is 0 (by construction) while $[\mathbf{1}_{(0,\mathbf{n})}]_{\text{boundary}} = 1$, forcing $c_i = 0$. If Ω_i has no boundary port, $\boldsymbol{\lambda}_i^u = c_i \mathbf{1}_{(0,\mathbf{n})}$ restricted to internal interfaces.

Interface continuity propagates $c_i = c_j$ for adjacent pores. Condition (A1) ensures every connected component contains a boundary pore where $c = 0$; hence $c_i = 0$ for all i and $\boldsymbol{\lambda} = \mathbf{0}$. Thus $\mathbf{S} \succ 0$.

Mass conservation. From $B_i \mathbf{U}_i = \mathbf{0}$ and the constant test function, $\sum_r M_{i,r}^{(0,\mathbf{n})} = 0$. The Schur assembly enforces continuity of $M^{(0,\mathbf{n})}$ on each internal interface; summing over pores gives global balance. $\square$

Remark 2.6 (Floating basins and coercivity). Condition (A2) fails for pores that lack a solid-wall facet (floating basins). In the watershed partition, such basins are merged into a neighbour with a larger shared interface area before the Schur assembly. This merge restores local coercivity (condition A2), but does not supply the global anchoring condition (A1), which is ensured separately by the inlet/outlet boundary-pore connections. The two conditions are distinct: (A2) guarantees K_i is invertible; (A1) eliminates the global constant-pressure kernel of $\mathbf{S}$.

A Error estimate as a Galerkin projection

We characterise the Affine-DDPNM as a Galerkin restriction of a non-overlapping domain-decomposition formulation with **broken** pressure ($Q_h^{\text{br}} = \bigoplus_i Q_{i,h}$). This C-DD reference differs

from the monolithic continuous- P_1 Taylor–Hood FEM. The error estimate below refers to the broken-pressure C-DD solution; its relation to the monolithic FEM is discussed in Remark A.5.

A.1 C-DD formulation (broken pressure)

On interface $\gamma_{i,r}$, the traction $\boldsymbol{\sigma}\mathbf{n}_{i,r}$ is expanded in the FE face basis $\{\psi_\ell\}$ with nodal coefficients $\boldsymbol{\eta}_{i,r}, \boldsymbol{\tau}_{i,r}^1, \boldsymbol{\tau}_{i,r}^2 \in \mathbb{R}^{m_{i,r}}$. The full C-DD traction is $\boldsymbol{\xi}_{i,r} = (\boldsymbol{\eta}_{i,r}; \boldsymbol{\tau}_{i,r}^1; \boldsymbol{\tau}_{i,r}^2) \in \mathbb{R}^{3m_{i,r}}$.

The Neumann coupling blocks are

$$\begin{aligned} [N_{i,r}^{\mathbf{n}}]_{\ell,j} &= \int_{\gamma_{i,r}} \psi_\ell(\phi_j^{(i)} \cdot \mathbf{n}_{i,r}) \, dS, \\ [N_{i,r}^{\mathbf{t}^\nu}]_{\ell,j} &= \int_{\gamma_{i,r}} \psi_\ell(\phi_j^{(i)} \cdot \mathbf{t}_{i,r}^\nu) \, dS \quad (\nu = 1, 2). \end{aligned}$$

Stacking all interfaces **vertically**,

$$N_i = \begin{bmatrix} N_i^{\mathbf{n}} \\ N_i^{\mathbf{t}_1} \\ N_i^{\mathbf{t}_2} \end{bmatrix} \in \mathbb{R}^{3 \sum_r m_{i,r} \times n_{i,u}}.$$

The local C-DD system is $K_i[V_i; P_i] = -[N_i^\top \boldsymbol{\xi}_i; \mathbf{0}] + (\text{boundary})$, yielding the local C-DD Schur complement $\hat{\mathbf{S}}_i = N_i H_i N_i^\top$. Global Boolean assembly with maps $\hat{R}_i^u$ gives

$$\hat{\mathbf{S}} \boldsymbol{\xi}^{\text{FE}} = \hat{\mathbf{F}}, \quad \hat{\mathbf{S}} = \sum_i (\hat{R}_i^u)^\top \hat{\mathbf{S}}_i \hat{R}_i^u. \quad (9)$$

Tangential sign convention for C-DD. Unlike the normal direction (where $\mathbf{n}_{j,r} = -\mathbf{n}_{i,r}$ is automatic), the tangential vectors $\mathbf{t}_{i,r}^\nu$ must have their relative orientation explicitly fixed. We adopt the same side-sign convention as the affine basis: $\mathbf{t}_{j,r}^\nu = -\mathbf{t}_{i,r}^\nu$ for $\nu = 1, 2$. The Boolean maps $\hat{R}_i^u$ incorporate a factor -1 for tangential entries of the second pore, ensuring that a single global tangential coefficient produces opposite-signed local tractions. This convention is implicit in the standard C-DD formulation (where tangential tractions are typically set to zero); making it explicit is necessary when tangential modes are retained.

A.2 Affine restriction operator

For interface γ_k with face mass matrix M_k , define $\mathbf{b}_k^{\alpha,\beta} \in \mathbb{R}^{m_k}$ by $(\mathbf{b}_k^{\alpha,\beta})_\ell = \int_{\gamma_k} \varphi_\alpha \psi_\ell \, dS$ and $P_k^{\alpha,\beta} = M_k^{-1} \mathbf{b}_k^{\alpha,\beta}$. Because the $[P_2]^3$ velocity trace on a triangular face exactly contains the affine functions $\{1, s, t\}$, the L^2 -projection is exact: $\sum_\ell P_k^{\alpha,\beta} \ell \psi_\ell = \varphi_\alpha$ on γ_k .

For each mode (α, β) , the **single non-zero block-row** is determined by β :

- $\beta = \mathbf{n}$: only the first (normal) block-row is non-zero;
- $\beta = \mathbf{t}_1$: only the second block-row is non-zero;
- $\beta = \mathbf{t}_2$: only the third block-row is non-zero.

The local restriction matrix $P_k^{\text{aff}} \in \mathbb{R}^{3m_k \times 9}$ is

$$P_k^{\text{aff}} = \begin{bmatrix} P_k^{0,\mathbf{n}} & \mathbf{0} & \mathbf{0} & P_k^{s,\mathbf{n}} & \mathbf{0} & \mathbf{0} & P_k^{t,\mathbf{n}} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & P_k^{0,\mathbf{t}_1} & \mathbf{0} & \mathbf{0} & P_k^{s,\mathbf{t}_1} & \mathbf{0} & \mathbf{0} & P_k^{t,\mathbf{t}_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & P_k^{0,\mathbf{t}_2} & \mathbf{0} & \mathbf{0} & P_k^{s,\mathbf{t}_2} & \mathbf{0} & \mathbf{0} & P_k^{t,\mathbf{t}_2} \end{bmatrix}. \quad (10)$$

The global restriction is $\mathcal{P}_{\text{aff}} = \bigoplus_{k=1}^m P_k^{\text{aff}} : \mathbb{R}^{9m} \rightarrow \mathbb{R}^{m_\Gamma}$.

A.3 Equivalence lemma

Lemma A.1 (Equivalence of assembled and projected systems). $\mathbf{S} = \mathcal{P}_{\text{aff}}^\top \hat{\mathbf{S}} \mathcal{P}_{\text{aff}}$ and $\mathbf{F} = \mathcal{P}_{\text{aff}}^\top \hat{\mathbf{F}}$.

Proof. For a pore i and internal interface γ , the primitive load vector (3) for mode (α, β) equals

$$\mathbf{f}_i^{(\gamma, \alpha, \beta)} = (N_{i, \gamma}^\beta)^\top P_{i, \gamma}^{\alpha, \beta},$$

where $N_{i, \gamma}^\beta$ is the coupling block for direction β . (The C-DD right-hand side is $-N_i^\top \boldsymbol{\xi}_i$; with the affine convention $-\sigma n = \sum \lambda \varphi \mathbf{d}$, we have $\boldsymbol{\xi}_i = -\mathcal{P}_{\text{aff}} \boldsymbol{\lambda}$ in the C-DD nodal basis, so the double minus cancels, yielding the positive relation above.)

The matrix of all internal primitive loads is $L_i^u = (N_i^u)^\top P_i^{\text{aff}}$. Hence $\mathbf{G}_i^{uu} = (L_i^u)^\top H_i L_i^u = (P_i^{\text{aff}})^\top \hat{\mathbf{S}}_i^{uu} P_i^{\text{aff}}$. Summing with the Boolean maps R_i^u and $\hat{R}_i^u$ and using the commutative relation $\hat{R}_i^u P_i^{\text{aff}} = \mathcal{P}_{\text{aff}}|_{\Omega_i} R_i^u$ gives the result. The forcing term follows analogously. $\square$

A.4 Error estimate and discussion

Theorem A.2 (Error estimate). *Let $\boldsymbol{\xi}^{FE}$ solve (9). Let $\boldsymbol{\lambda}^{\text{aff}}$ solve $\mathbf{S} \boldsymbol{\lambda}^{\text{aff}} = \mathbf{F}$ and $\boldsymbol{\xi}^{\text{aff}} = \mathcal{P}_{\text{aff}} \boldsymbol{\lambda}^{\text{aff}}$. Assume $\hat{\mathbf{S}} \succeq 0$ on the full C-DD trace space and that its kernel is eliminated by boundary anchoring (as for $\mathbf{S}$ in Theorem 2.5, but on the C-DD level). Then $\|\cdot\|_{\hat{\mathbf{S}}}$ is a norm on the quotient space modulo $\ker(\hat{\mathbf{S}})$, and*

$$\|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{\text{aff}}\|_{\hat{\mathbf{S}}} = \min_{\boldsymbol{\mu} \in \mathbb{R}^{9m}} \|\boldsymbol{\xi}^{FE} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\hat{\mathbf{S}}}, \quad (11)$$

$$\sum_i \|\mathbf{U}_i^{FE} - \mathbf{U}_i^{\text{aff}}\|_{A_i}^2 = \|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{\text{aff}}\|_{\hat{\mathbf{S}}}^2. \quad (12)$$

Proof. Lemma A.1 reduces the Affine system to the Galerkin restriction. Galerkin orthogonality and the Pythagorean identity in the $\hat{\mathbf{S}}$ -semi-inner-product yield (11). Equation (12) follows from the C-DD energy identity. The equality case $\boldsymbol{\mu} = \boldsymbol{\lambda}^{\text{aff}}$ holds by definition of $\boldsymbol{\xi}^{\text{aff}}$; injectivity of $\mathcal{P}_{\text{aff}}$ is not required for the equality case (it guarantees uniqueness of the representing coefficient). $\square$

Remark A.3 (Enrichment guarantee). $\text{Range}(\mathcal{P}_{P0}) \subset \text{Range}(\mathcal{P}_{\text{aff}})$, so the affine error in $\|\cdot\|_{\hat{\mathbf{S}}}$ never exceeds the classic error. This monotonicity is restricted to the $\hat{\mathbf{S}}$ -energy (semi-)norm and does not automatically extend to the velocity L^2 , pressure L^2 , broken H^1 , or outlet flux. The 73–90% L^2 reductions observed numerically are consistent with the energy-norm enrichment, not formally implied by it.

Remark A.4 (Mesh dependence). Equation (11) is a discrete optimality statement for a fixed mesh. A continuum $h \rightarrow 0$ limit requires a fixed interface topology, shape-regular refinement with stable trace spaces, and convergence of $\hat{\mathbf{S}}_h$. Watershed meshes do not satisfy the fixed-topology requirement (basin count varies with mesh size); the observed h -insensitivity is therefore a numerical observation, not the consequence of a limiting argument.

Remark A.5 (Broken-pressure C-DD versus monolithic FEM). The broken-pressure C-DD formulation uses $Q_h^{br} = \bigoplus_i Q_{i,h}$, whereas the monolithic Taylor–Hood FEM uses $Q_h^c \subset H^1(\Omega)$. These are different discrete problems. The “Exact-FE-Schur” computation in the companion code compares the primal Schur complement of the continuous-pressure global system against the direct monolithic solve; their agreement to $O(10^{-12})$ verifies the block-elimination algebra of that single global matrix. It does not constitute a comparison between the broken-pressure C-DD formulation and the monolithic FEM. Whether the two reference systems produce identical velocity solutions for the tested geometries is an open question; Theorem A.2 is a statement about the broken-pressure C-DD reference only.

References

- [1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.